Software Test Plan (STP)

Project: Online Exam Portal (OEP)
Team Number: 9 | Section: A
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Revision History

Version	Date	Author	Change Summary	Approval
1.0	16-09-2026	Aadish Sarin	Initial Software Test Plan derived from SRS v1.0 and SAD v1.0	Pending

Team & Testing Work Allocation

Testing work is divided equally. Every member writes and executes the test cases for a module he did NOT develop (cross-testing), so that no defect is hidden by developer bias, and every member reviews one other member's test cases.

SRN	Member	Module Developed	Module Tested (cross)	Test Case IDs Owned	Extra Duty
PES2UG24CS002	A S Vincent	M1 - Authentication & User/Role Mgmt	M4 - Evaluation & Analytics	TC-Eval-01..04 + variants	Security test execution (TC-Sec-*)
PES2UG24CS007	Aadish Sarin (Lead)	M2 - Exam & Question-Bank Mgmt	M1 - Authentication & User/Role Mgmt	TC-Auth-01..04 + variants	QA Lead: plan, RTM, defect triage, sign-off
PES2UG24CS034	Aditya Rao	M3 - Exam Delivery / Test-Taking	M2 - Exam & Question-Bank Mgmt	TC-Exam-01..04 + variants	Test environment & seed data setup
PES2UG24CS045	Akshansh Agrawal	M4 - Evaluation, Results & Analytics	M3 - Exam Delivery / Test-Taking	TC-Take-01..04 + variants	Performance (JMeter) & usability testing

Table of Contents

1. Introduction 2. Test Items 3. Features to be Tested 4. Features Not to be Tested 5. Test Approach / Strategy 5.5 Security Validation 6. Test Environment 7. Test Schedule 8. Test Deliverables 9. Roles and Responsibilities 10. Risks and Mitigation 11. Assumptions & Dependencies 12. Suspension & Resumption Criteria 13. Test Case Management & Traceability 14. Test Metrics & Reporting 15. Approvals Annexure A - Detailed Test Cases Annexure B - Defect Report Template Annexure C - Full RTM

1. Introduction

1.1 Purpose

This Software Test Plan (STP) defines the scope, approach, resources, schedule and exit criteria for testing the Online Exam Portal (OEP). It establishes how each of the 26 requirements recorded in the SRS (16 functional, 5 non-functional, 5 security) will be verified, who verifies it, in which environment, and what evidence is produced. The document is the agreed contract between the four developers of Team 9 and the course evaluator on what "tested" means for this project.

1.2 Scope

In scope: functional, integration, system, performance, security, usability and regression testing of all four OEP modules - M1 Authentication & User/Role Management, M2 Exam & Question-Bank Management, M3 Exam Delivery / Test-Taking Engine, and M4 Evaluation, Results & Analytics - together with the REST API layer and the relational database that back them.

Out of scope: penetration testing by an external agency, testing of third-party infrastructure (browser internals, the email/SMS gateway service itself, cloud provider uptime), and mobile-native application testing, since the OEP is delivered as a responsive web application only.

1.3 References

Ref	Document	Version / Date
R1	Software Requirements Specification - Online Exam Portal (Team 9)	v1.0, 06-09-2026
R2	Software Architecture Document - Online Exam Portal (Team 9)	v1.0, 14-09-2026
R3	Use-Case Diagrams - Online Exam Portal (Team 9), 2 diagrams	06-09-2026
R4	Team 9 Kickoff & Task Assignment (sprint plan, Definition of Done)	14-09-2026
R5	Mini Project Test Plan Template for SE (course-provided)	Current
R6	IEEE 829 - Standard for Software Test Documentation	Reference standard

1.4 Definitions and Acronyms

Term	Meaning
OEP	Online Exam Portal - the system under test
STP	Software Test Plan (this document)
SUT	System Under Test
RTM	Requirements Traceability Matrix - links each requirement to its test case(s) and status
FR / NFR / SR	Functional / Non-Functional / Security Requirement (IDs OEP-F-###, OEP-NF-###, OEP-SR-###)
TC	Test Case (IDs TC-Auth-##, TC-Exam-##, TC-Take-##, TC-Eval-##, TC-Perf-##, TC-Sec-##, TC-Rel-##, TC-UX-##)

RBAC	Role-Based Access Control
UAT	User Acceptance Testing
Defect Severity	Critical / High / Medium / Low - impact of the defect on the system
Defect Priority	P1 / P2 / P3 - urgency with which the defect must be fixed
Regression	Re-running previously passed tests to confirm new changes broke nothing
Smoke Test	Minimal test set proving the build is stable enough to test further

2. Test Items

The following build artifacts of the Online Exam Portal are the items under test:

#	Test Item	Module	Owner (Developer)	Interface Tested
1	Registration, Login, Password-Reset and RBAC middleware	M1	A S Vincent	Web UI + /api/auth/register, /api/auth/login
2	Exam CRUD, Question-Bank CRUD, scheduling, randomisation	M2	Aadish Sarin	Instructor UI + /api/exams/*
3	Exam listing, timer, auto-save, auto-submit, question navigation	M3	Aditya Rao	Student exam screen + /api/exams/{examId}/attempt, /api/attempts/{attemptId}/answers
4	Auto-grading, manual grading, result publication, analytics reports	M4	Akshansh Agrawal	Instructor/Student UI + /api/results/*
5	Relational database schema (users, exams, questions, attempts, answers, results, audit_log)	Shared	All members	SQL / ORM layer
6	Cross-cutting: HTTPS/TLS configuration, session/JWT handling, audit logging	Shared	Vincent + Aadish	Transport & middleware

3. Features to be Tested

Every requirement in the SRS is mapped to at least one test case. Test priority is inherited from the requirement priority.

3.1 Functional Features

Req ID	Feature to be Tested	Module	Priority	Test Case(s)
OEP-F-001	User registration with unique email and validated password	M1	High	TC-Auth-01, TC-Auth-01a, TC-Auth-01b
OEP-F-002	Login with email/password and role-based session creation	M1	High	TC-Auth-02, TC-Auth-02a
OEP-F-003	Password reset via OTP / reset link	M1	Medium	TC-Auth-03, TC-Auth-03a
OEP-F-004	Role-based access control (Student / Instructor / Admin)	M1	High	TC-Auth-04, TC-Auth-04a
OEP-F-005	Create, edit and delete an exam (instructor)	M2	High	TC-Exam-01, TC-Exam-01a
OEP-F-006	Add questions (MCQ, True/False, short answer) to question bank	M2	High	TC-Exam-02, TC-Exam-02a
OEP-F-007	Set exam schedule, duration and total marks	M2	High	TC-Exam-03, TC-Exam-03a
OEP-F-008	Randomise / shuffle question order per student	M2	Medium	TC-Exam-04

OEP-F-009	List available / scheduled exams for an eligible student	M3	High	TC-Take-01, TC-Take-01a
OEP-F-010	Present exam with server-enforced countdown timer	M3	High	TC-Take-02, TC-Take-02a
OEP-F-011	Periodic auto-save of answers and auto-submit at time-up	M3	High	TC-Take-03, TC-Take-03a
OEP-F-012	Question navigation and mark-for-review	M3	Medium	TC-Take-04
OEP-F-013	Auto-grade objective questions against the answer key	M4	High	TC-Eval-01, TC-Eval-01a
OEP-F-014	Manual grading interface for subjective answers	M4	High	TC-Eval-02
OEP-F-015	Publish results; student views own scorecard only	M4	High	TC-Eval-03, TC-Eval-03a
OEP-F-016	Analytics / reports (average, pass %, item analysis)	M4	Medium	TC-Eval-04

3.2 Non-Functional Features

Req ID	Feature to be Tested	Category	Priority	Test Case(s)
OEP-NF-001	Exam page / navigation load time <= 3 s at 90th percentile	Performance	High	TC-Perf-01
OEP-NF-002	500 concurrent students without failure or data loss	Scalability	High	TC-Perf-02
OEP-NF-003	99.5% availability in exam window; answers restored after reconnect	Reliability	High	TC-Rel-01
OEP-NF-004	Credentials hashed; all traffic over TLS 1.2+	Security	High	TC-Sec-01
OEP-NF-005	Responsive UI, WCAG 2.1 AA conformance where applicable	Usability	Medium	TC-UX-01

3.3 Security Features

Req ID	Feature to be Tested	Security Objective	Priority	Test Case(s)
OEP-SR-001	TLS 1.2+ mandatory; plain HTTP redirected to HTTPS	SO-1	High	TC-Sec-02
OEP-SR-002	Passwords stored with salted bcrypt/argon2 hash	SO-1	High	TC-Sec-03
OEP-SR-003	Server-side RBAC on every protected API endpoint	SO-1	High	TC-Sec-04
OEP-SR-004	Questions hidden before start; correct answers never in client payload	SO-2	High	TC-Sec-05
OEP-SR-005	Audit log of login, submission and grading actions	SO-2	Medium	TC-Sec-06
(cross-cutting)	Input validation against SQL injection and XSS	SO-1, SO-2	High	TC-Sec-07

4. Features Not to be Tested

Feature / Area	Reason for Exclusion
Internal behaviour of the third-party email / SMS gateway used for OTP	External service owned by the provider; only the OEP integration point (request sent, OTP accepted) is tested via TC-Auth-03.
Browser rendering engine internals and browser-specific CSS bugs	Testing is limited to the supported browser matrix (Chrome, Edge, Firefox - latest two versions); vendor defects are out of team control.
Native mobile applications (Android / iOS)	The OEP scope is a responsive web application only; no native app exists in this release.

Database engine internals, backup/restore and disaster recovery drills	Infrastructure/DBA responsibility; not part of the academic mini-project scope or lab environment.
Full external penetration testing and formal security certification	Requires specialised tooling, authorisation and time beyond the lab scope. Basic security validation is covered in Section 5.1.
Proctoring via webcam / face recognition	Listed as a future enhancement in the SRS; not implemented in this release, therefore nothing to verify.
Load beyond 500 concurrent users	OEP-NF-002 caps the stated capacity at 500; higher loads are not a committed requirement.

5. Test Approach / Strategy

Testing follows a bottom-up progression: each developer unit-tests his own module, the cross-assigned tester then verifies that module functionally, and the whole team performs integration, system, performance, security and acceptance testing on the merged build. Testing is a mix of manual (UI, usability, exploratory) and automated (unit, API, load) techniques.

5.1 Test Levels

Level	What is Verified	Technique / Tool	Performed By	When
Unit Testing	Individual functions, service methods, validators and grading logic in isolation	Jest / Mocha, mocked DB	Developer of the module	Continuously, before every merge to dev
Integration Testing	Module-to-module and UI-to-API contracts, e.g. M2 exam -> M3 delivery -> M4 grading	Postman collections, supertest	Cross-assigned tester	After each module merge
System Testing	Complete end-to-end business flows across all four modules on the integrated build	Manual scripted tests + Selenium/Playwright	Whole team, coordinated by QA Lead	System test window
Performance Testing	Response time (NF-001) and 500-user concurrency (NF-002)	Apache JMeter / k6	Akshansh Agrawal	After system test passes
Security Testing	TLS, hashing, RBAC, answer-key leakage, audit log, injection/XSS	Postman, browser dev-tools, OWASP ZAP (baseline), SQL inspection	A S Vincent + Aadish Sarin	After system test passes
Regression Testing	Previously passed test cases re-run after every defect fix	Automated suite + smoke checklist	Cross-assigned tester	After each defect fix batch
User Acceptance Testing	Requirement acceptance criteria validated from the user's point of view	Manual, scenario-based	Aadish Sarin (acting Product Owner) + peer reviewers	Final week before sign-off

5.2 Test Types

Test Type	Objective	Coverage in OEP
Functional	Confirm each FR behaves per its acceptance criteria	All 16 FRs - positive paths plus negative/boundary paths where applicable
Negative / Boundary	Confirm invalid input is rejected gracefully	Duplicate email, weak password, expired OTP, past exam date, zero-mark exam, submission after time-up
Integration	Confirm data flows correctly	Exam created (M2) appears to student (M3);

	between modules	submission (M3) grades in (M4)
Regression	Confirm fixes do not break working features	Smoke suite of 12 High-priority cases re-run on every build
Performance / Load	Confirm NF-001 and NF-002 under realistic load	500 virtual users, 3-second 90th-percentile threshold
Reliability / Recovery	Confirm answers survive network loss and refresh	TC-Rel-01: disconnect mid-exam, reconnect, verify answers restored
Usability / Accessibility	Confirm UI is clear and WCAG 2.1 AA conformant	Keyboard navigation, colour contrast, screen-reader labels, responsive breakpoints
Security	Confirm confidentiality, integrity and auditability	SO-1 and SO-2 objectives via TC-Sec-01..07
Compatibility	Confirm behaviour across supported browsers/resolutions	Chrome, Edge, Firefox at 1920x1080, 1366x768 and tablet width

5.3 Entry Criteria

- SRS v1.0 and SAD v1.0 are baselined and approved; no open High-severity requirement ambiguity.
- This Test Plan is reviewed and approved by the Team Lead and Course Coordinator.
- Test cases for the feature under test are written, peer-reviewed and stored in the shared test repository.
- A stable build is deployed to the test environment and passes the 12-case smoke suite.
- Test environment (application server, database, seed data) is available and reset to a known state.
- Seed/dummy data script has been executed so users, exams, questions and attempts exist.
- Defect tracking board (Jira) is configured with the agreed severity/priority scheme.

5.4 Exit Criteria

- 100% of planned test cases for all High-priority requirements have been executed.
- Test case pass rate >= 95% overall, and 100% for High-priority functional and security test cases.
- Zero open Critical defects and zero open High-severity defects.
- Open Medium/Low defects are documented, triaged and formally accepted as known issues by the Team Lead.
- All five NFRs meet their measured thresholds (<=3 s at p90, 500 concurrent users, >=99.5% uptime in the window, TLS+hashing verified, accessibility audit passed).
- RTM shows every one of the 26 requirements as Accepted (status A) with linked evidence.
- Regression suite passes on the final release candidate build.
- Test Summary Report is produced and signed off by the QA Lead, Dev Lead and Product Owner.

5.5 Security Validation

Security validation verifies the two security objectives declared in the SRS:

- SO-1 - Confidentiality and integrity of exam content and results: only authorised users reach exams, answers and marks.
- SO-2 - Prevention of malpractice and tampering, with a trustworthy audit trail of submissions and grading.

Security Check	Method	Expected Result	Req ID	TC
Transport encryption	Inspect the TLS handshake in browser dev-tools; attempt plain HTTP request	TLS 1.2 or higher negotiated; HTTP request 301-redirected to HTTPS	OEP-SR-001, OEP-NF-004	TC-Sec-01, TC-Sec-02
Password storage	Register a user, then read the users table directly	Stored value is a salted bcrypt/argon2 hash; plaintext password appears nowhere	OEP-SR-002	TC-Sec-03
Authorisation bypass	Call an instructor-only API using a valid Student token (Postman)	HTTP 403 Forbidden; no data returned; attempt written to audit log	OEP-SR-003, OEP-F-004	TC-Sec-04
Answer-key leakage	Open an exam and inspect the network payload and page source	No isCorrect / correctOption field present in any client-bound response	OEP-SR-004	TC-Sec-05
Pre-start exam exposure	Request an exam before its scheduled start time	Questions not served; error/locked message returned	OEP-SR-004	TC-Sec-05
Audit logging	Perform login, submission and grading, then query audit_log	One immutable record per action with actor, action, entity and timestamp	OEP-SR-005	TC-Sec-06
Injection / XSS	Submit '" OR 1=1 --' and '<script>alert(1)</script>' into login and answer fields	Input rejected or escaped; no SQL error, no script execution	Cross-cutting	TC-Sec-07
Session handling	Log out, then replay the old token; leave a session idle past timeout	Token invalidated; expired session forces re-login	OEP-F-002	TC-Auth-02a

6. Test Environment

6.1 Hardware

Component	Specification	Purpose
Developer / Tester workstations (x4)	Windows 11, Intel i5 or better, 8-16 GB RAM, SSD	Test design, manual execution, defect logging
Application server (local or cloud VM)	Linux, 2 vCPU, 4 GB RAM minimum	Hosts the Node.js/Express API build under test
Database server	Same VM or separate instance, 2 GB RAM	MySQL/PostgreSQL instance with the test schema
Load generation machine	One workstation running JMeter/k6 with 8 GB RAM	Generates 500 virtual users for TC-Perf-02
Network	Campus LAN / Wi-Fi, with a throttling profile for recovery testing	Simulates real student connectivity and disconnection

6.2 Software

Layer	Software / Version
Client	Google Chrome, Microsoft Edge, Mozilla Firefox - latest two stable versions
Front end	React (Vite build), responsive CSS
Back end	Node.js with Express, REST/JSON over HTTPS
Database	MySQL 8.x or PostgreSQL 15.x
Security libraries	bcrypt / argon2 for hashing, JWT for sessions, TLS 1.2+ certificate
Operating systems	Windows 11 (clients), Linux (server)

6.3 Tools

Purpose	Tool	Used By
Unit testing	Jest / Mocha + Chai	Each developer on his own module
API and integration testing	Postman (shared collection) + Newman for CLI runs	Cross-assigned testers
UI / end-to-end automation	Selenium WebDriver or Playwright	Aditya Rao, Akshansh Agrawal
Performance / load testing	Apache JMeter or k6	Akshansh Agrawal
Security scanning	OWASP ZAP (baseline scan), browser dev-tools, SQL client	A S Vincent
Accessibility audit	Lighthouse / axe DevTools	Akshansh Agrawal
Defect tracking	Jira (Team 9 board) with Bug issue type	All members
Version control / CI	Git + GitHub, branch protection on main	All members
Test documentation	MS Word (this STP), shared spreadsheet for execution log	Aadish Sarin (QA Lead)

6.4 Test Data

Test data is produced by a single seed script (seed.sql / seed.js) checked into the repository, so that every tester works against an identical, resettable dataset. This also removes the dependency chain that would otherwise force M3 and M4 to wait for M1 and M2 to finish.

Data Set	Contents	Used For
Users	3 Students, 2 Instructors, 1 Admin with known credentials; 1 locked/invalid account	Login, RBAC, negative authentication tests
Question bank	30 questions: 20 MCQ, 5 True/False, 5 short answer, with answer keys	Exam creation, randomisation, auto-grading, manual grading
Exams	1 upcoming (not started), 1 live (in window), 1 completed, 1 unpublished	Exam listing eligibility, pre-start lock, result publication
Attempts	2 completed attempts with known expected scores, 1 in-progress attempt	Auto-grade verification, analytics, auto-save/recovery
Boundary data	Empty strings, 256-character inputs, duplicate email, expired OTP, 0 and max marks	Negative and boundary value testing
Malicious data	SQL injection and XSS payload strings	TC-Sec-07 input validation testing

7. Test Schedule

Dates align with the sprint plan in the Team 9 Kickoff document. Each milestone has a named owner and a concrete exit artifact.

#	Milestone / Activity	Start	End	Owner	Exit Artifact
1	Test Plan review and approval	16-09-2026	17-09-2026	Aadish Sarin	Approved STP v1.0
2	Test case design (all modules, cross-assigned)	17-09-2026	19-09-2026	All members	Annexure A test cases
3	Peer review of test cases	19-09-2026	20-09-2026	Aadish Sarin	Review comments closed
4	Test environment setup + seed data script	19-09-2026	20-09-2026	Aditya Rao	Reproducible test environment
5	Unit testing (per module)	21-09-2026	23-09-2026	Each developer	Unit test reports, coverage summary
6	Integration testing (M1-M2-M3-	23-09-	24-09-	Cross-assigned	Postman run results

	M4 + API)	2026	2026	testers	
7	System testing (end-to-end flows)	25-09-2026	28-09-2026	Whole team	System test execution log
8	Performance testing (NF-001, NF-002)	29-09-2026	29-09-2026	Akshansh Agrawal	JMeter report with p90 latency
9	Security validation (SR-001..005, TC-Sec-07)	29-09-2026	30-09-2026	A S Vincent	Security test evidence pack
10	Usability / accessibility audit (NF-005)	30-09-2026	30-09-2026	Akshansh Agrawal	Lighthouse/axe report
11	Defect fixing and retesting	01-10-2026	02-10-2026	Developers + testers	Closed defect list
12	Regression testing on release candidate	02-10-2026	03-10-2026	Cross-assigned testers	Regression pass report
13	User Acceptance Testing	03-10-2026	03-10-2026	Aadish Sarin (PO role)	Signed UAT checklist
14	Test Summary Report and final sign-off	04-10-2026	04-10-2026	Aadish Sarin	Test Summary Report + completed RTM

8. Test Deliverables

#	Deliverable	Description	Owner	Due
1	Test Plan (this document)	Scope, strategy, schedule, criteria and responsibilities for all OEP testing	Aadish Sarin	17-09-2026
2	Test Cases (Annexure A)	Approximately 40 test cases covering all 26 requirements, with steps, test data and expected results	All members (cross-assigned)	19-09-2026
3	Test Scripts	Jest/Mocha unit specs, Postman collection, Selenium/Playwright scripts, JMeter test plan	Each developer + Akshansh	23-09-2026
4	Test Data / Seed Script	seed.sql or seed.js creating users, questions, exams, attempts and boundary data	Aditya Rao	20-09-2026
5	Test Execution Log	Per-cycle record of TC ID, build, tester, date, actual result, pass/fail, defect ID	Cross-assigned testers	Rolling
6	Defect Reports (Annexure B)	Jira bug tickets using the standard template with severity, priority, steps and evidence	Whoever finds the defect	Rolling
7	Requirements Traceability Matrix (Annexure C)	All 26 requirements traced to design spec, module, test case and final status	Aadish Sarin	04-10-2026
8	Test Summary Report	Cases planned vs executed vs passed, defect metrics, NFR measurements, exit criteria verdict, recommendation	Aadish Sarin	04-10-2026

9. Roles and Responsibilities

Role	Name (SRN)	Responsibility
QA Lead / Test Manager	Aadish Sarin (PES2UG24CS007)	Owns this Test Plan; allocates test cases; runs defect triage; maintains the RTM; verifies exit criteria; writes the Test Summary Report; final sign-off.
Test Engineer - Authentication (M1)	Aadish Sarin (PES2UG24CS007)	Designs and executes TC-Auth-01..04 and their negative variants; validates RBAC behaviour; reports M1 defects to A S Vincent.
Test Engineer - Exam & Question Bank (M2)	Aditya Rao (PES2UG24CS034)	Designs and executes TC-Exam-01..04; verifies scheduling, marks and randomisation; also builds and maintains the test environment and seed data.
Test Engineer - Exam	Akshansh Agrawal	Designs and executes TC-Take-01..04 and TC-Rel-01; runs

Delivery (M3)	(PES2UG24CS045)	performance tests TC-Perf-01/02 and the accessibility audit TC-UX-01.
Test Engineer - Evaluation & Analytics (M4)	A S Vincent (PES2UG24CS002)	Designs and executes TC-Eval-01..04; verifies grading accuracy and report values; additionally executes security tests TC-Sec-01..07.
Developer - M1 Authentication	A S Vincent (PES2UG24CS002)	Unit-tests own module; fixes assigned defects within the agreed turnaround; supports the M1 tester with logs and environment access.
Developer - M2 Exam & Question Bank	Aadish Sarin (PES2UG24CS007)	Unit-tests own module; fixes assigned defects; maintains API contract documentation used by testers.
Developer - M3 Exam Delivery	Aditya Rao (PES2UG24CS034)	Unit-tests own module; fixes assigned defects; supports timer and auto-save investigation.
Developer - M4 Evaluation & Analytics	Akshansh Agrawal (PES2UG24CS045)	Unit-tests own module; fixes assigned defects; provides expected-score calculations for grading verification.
Product Owner (acting)	Aadish Sarin (PES2UG24CS007)	Confirms acceptance criteria, prioritises defects for fixing, approves UAT results and accepts known issues.
Course Coordinator / Evaluator	Course Faculty	Reviews and approves the Test Plan and the Test Summary Report; evaluates the deliverables.

Cross-testing rule: no member tests the module he wrote. This removes developer bias, forces the API contract and the documentation to be good enough for someone else to use, and keeps the four workloads equal - each member writes one module, tests one module, and owns one cross-cutting specialism (security, performance, environment, or QA management).

10. Risks and Mitigation

#	Risk	Impact	Likelihood	Mitigation
R1	A module is delivered late, blocking the cross-assigned tester	High	Medium	Seed data and API stubs let testing start against mocks; weekly stand-up tracks module readiness; QA Lead reassigns test effort.
R2	Test environment differs from the environment used for demo/evaluation	High	Medium	Pin versions in package.json; document setup in README; use the same seed script everywhere.
R3	Load test of 500 concurrent users cannot be generated on student hardware	Medium	High	Run a scaled test (e.g. 100 users) and extrapolate, or use a cloud k6 free tier; document the deviation in the Test Summary Report.
R4	Timer and auto-save defects appear only under poor network conditions	High	Medium	Explicitly test with browser network throttling and forced disconnects (TC-Rel-01); enforce the timer server-side, never client-side.
R5	Requirements change after test cases are written	Medium	Medium	Any SRS change raises a new version and triggers an RTM impact review; affected test cases are updated before execution.
R6	Defects are found late because integration happens too late	High	Medium	Walking-skeleton build integrated in Sprint 1; merge to dev at least twice a week; smoke suite runs on every merge.
R7	Security defect (answer key leaked in API response) discovered near submission	Critical	Low	TC-Sec-05 is executed as soon as the exam-delivery API exists, not at the end; payload inspection is part of the M3 Definition of Done.
R8	Team member unavailable due to illness or other coursework	Medium	Medium	Every test case is documented well enough for another member to execute; QA Lead

				maintains a backup owner per module.
R9	Inconsistent defect reporting makes triage slow	Low	High	Mandatory defect template (Annexure B); tickets missing steps or evidence are returned to the reporter.
R10	Data loss in the test database wipes evidence of executed tests	Medium	Low	Execution logs and screenshots stored in the shared repository, not only in the database; seed script allows fast rebuild.

11. Assumptions and Dependencies

11.1 Assumptions

- The SRS (v1.0) and SAD (v1.0) are stable baselines; any change is version-controlled and communicated to the QA Lead.
- All four members have working development environments and access to the shared Git repository and Jira board.
- The test environment mirrors the demo environment in versions and configuration.
- A valid TLS certificate (self-signed is acceptable for the lab) is available so TLS tests can be executed.
- The seed script is executed before each test cycle, so test results are reproducible.
- Test data is synthetic; no real student personal data is used at any point.
- Each member allocates the agreed hours per week to testing alongside development.

11.2 Dependencies

Dependency	Depends On	Impact if Delayed
Testing of M2 exam listing end-to-end	M1 login and RBAC (OEP-F-002, F-004)	Cannot authenticate as Instructor; mitigate with seeded session tokens
Testing of M3 exam delivery	M2 exams and questions (OEP-F-005..007)	No exam to take; mitigate with seeded exams in the database
Testing of M4 grading and analytics	M3 submitted attempts (OEP-F-011)	No submissions to grade; mitigate with seeded completed attempts
Security tests TC-Sec-01/02	HTTPS enabled on the test server	TLS verification blocked; must be configured before 29-09-2026
Performance tests TC-Perf-01/02	Stable integrated build plus load-generation tooling	Performance risk discovered too late to fix
OTP password-reset test TC-Auth-03	Email/SMS gateway credentials or a mail-catcher stub	Test blocked; use a local SMTP catcher such as MailHog
RTM completion	All test executions recorded with evidence	Cannot demonstrate exit criteria at sign-off

12. Suspension and Resumption Criteria

12.1 Suspension Criteria

Testing of the affected area is suspended and the build is rejected if any of the following occurs:

- The build fails the 12-case smoke suite (login, create exam, take exam, submit, view result cannot complete).
- A Critical defect blocks more than 30% of the planned test cases in the current cycle.
- The test environment or database is unavailable, or the seed data is corrupted.
- A security defect that exposes credentials, answer keys or another student's results is discovered - all testing of that module stops until it is fixed.
- Requirements for the feature under test are changed mid-cycle and test cases no longer match the SRS.
- The build under test does not match the version recorded in the execution log (build traceability lost).

12.2 Resumption Criteria

Testing resumes only when all of the following are true:

- The blocking defect is fixed, verified by the reporter, and closed in Jira.
- A new build is deployed and passes the full smoke suite.
- The test environment is restored and the seed script has been re-run to a known state.
- Affected test cases are updated and re-approved if requirements changed.
- The QA Lead records the suspension period, its cause and the resumption decision in the Test Summary Report, and re-runs the regression suite before normal testing continues.

13. Test Case Management and Traceability

Every requirement traces forward to at least one test case, and each test case may trace back to one or more requirements where the test verifies multiple requirements. A requirement is only marked Accepted (A) in the RTM when all of its linked test cases have passed on the release-candidate build and no open High or Critical defect is linked to it.

13.1 Identification Scheme

Artifact	ID Pattern	Example
Functional requirement	OEP-F-###	OEP-F-011 (auto-save and auto-submit)
Non-functional requirement	OEP-NF-###	OEP-NF-002 (500 concurrent users)
Security requirement	OEP-SR-###	OEP-SR-003 (server-side RBAC)
Design specification	DS-<Area>-##	DS-Take-02
Test case (positive)	TC-<Area>-##	TC-Take-03
Test case (negative variant)	TC-<Area>-##a / ##b	TC-Take-03a
Defect	OEP-BUG-###	OEP-BUG-014
Test cycle	CYCLE-##	CYCLE-02 (system test)

13.2 Status Codes Used in the RTM

Code	Meaning	Condition to Enter this State
N	Not started	Test case written but not yet executed against a build
P	Partial	Some test cases executed or passed, or a defect is still open against the requirement
A	Accepted	All linked test cases passed on the release candidate; no open High/Critical defect; evidence attached

13.3 Sample Traceability Rows

The complete matrix for all 26 requirements is given in Annexure C. A representative extract:

Req ID	Requirement (short)	Design Spec	Module	Test Case(s)	Status
OEP-F-002	Login and role-based session	DS-Auth-02	M1 AuthModule	TC-Auth-02, TC-Auth-02a	N
OEP-F-011	Auto-save and auto-submit	DS-Take-02	M3 DeliveryModule	TC-Take-03, TC-Take-03a, TC-Rel-01	N
OEP-F-013	Auto-grade objective questions	DS-Eval-01	M4 EvalModule	TC-Eval-01, TC-Eval-01a	N
OEP-NF-002	500 concurrent students	DS-Perf-02	All modules	TC-Perf-02	N
OEP-SR-004	No answer key in client payload	DS-Sec-03	M2 + M3	TC-Sec-05	N

14. Test Metrics and Reporting

14.1 Metrics Collected

Metric	Formula	Target	Reported By
Test case execution %	(Executed / Planned) x 100	100% for High priority	QA Lead, per cycle
Test case pass %	(Passed / Executed) x 100	>= 95% overall, 100% High priority	QA Lead, per cycle
Requirement coverage %	(Requirements with >=1 executed TC / 26) x 100	100%	QA Lead, from RTM
Defect density	Defects found / module (or per KLOC)	Tracked for comparison, no hard target	QA Lead
Defect severity distribution	Count by Critical / High / Medium / Low	0 open Critical or High at exit	QA Lead
Defect removal efficiency	(Defects found before UAT / Total defects) x 100	>= 90%	QA Lead at closure
Defect ageing	Average days from defect Open to Closed	<= 2 days for Critical/High	QA Lead
Defect rejection rate	(Rejected or duplicate defects / Total reported) x 100	<= 10% (indicates report quality)	QA Lead
Response time p90 (NF-001)	90th percentile page/API latency under load	<= 3 seconds	Akshansh Agrawal
Concurrency sustained (NF-002)	Max virtual users with 0 errors and no data loss	>= 500	Akshansh Agrawal
Availability (NF-003)	Uptime during simulated exam window	>= 99.5%	Akshansh Agrawal

14.2 Reporting Cadence

Report	Frequency	Audience	Content
Daily test status note	Daily during execution windows	Team 9 (stand-up)	Cases executed, passed, failed, blockers raised
Defect triage summary	Twice weekly	Team 9	New, open, fixed and reopened defects with assigned owners

Cycle test report	End of each test cycle	Team 9 + Course Coordinator	Metrics from 14.1, exit-criteria gap analysis
Test Summary Report	Once, at closure (04-10-2026)	Course Coordinator	Overall verdict, all metrics, known issues, completed RTM, recommendation to release

14.3 Defect Severity and Priority Scheme

Severity	Definition	Example in OEP	Target Fix Time
Critical	System unusable or data/security integrity lost	Correct answers visible in the API payload; submitted answers not saved	Same day
High	Major function fails with no workaround	Timer does not auto-submit; student sees another student's result	Within 1 day
Medium	Function fails but a workaround exists	Question randomisation does not shuffle; analytics average slightly off	Within 3 days
Low	Cosmetic or minor usability issue	Misaligned label, unclear error wording, missing tooltip	Before release if time permits

15. Approvals

By signing below, the approvers confirm that the scope, strategy, schedule and exit criteria defined in this Test Plan are understood and agreed, and that testing may commence.

Role	Name	Signature	Date
QA Lead / Test Manager	Aadish Sarin (PES2UG24CS007)		
Dev Lead	A S Vincent (PES2UG24CS002)		
Product Owner (acting)	Aadish Sarin (PES2UG24CS007)		
Team Member - Reviewer	Aditya Rao (PES2UG24CS034)		
Team Member - Reviewer	Akshansh Agrawal (PES2UG24CS045)		
Course Coordinator			

Annexure A - Detailed Test Cases

Column key: TC ID | Requirement traced | Test scenario | Pre-conditions | Test steps | Test data | Expected result | Type | Priority | Tester. Actual result and Pass/Fail are filled during execution in the Test Execution Log.

A.1 Module M1 - Authentication & User/Role Management (Tester: Aadish Sarin)

TC ID	Req ID	Test Scenario	Pre-conditions	Test Steps	Test Data	Expected Result	Type	Pri	Tester
TC-Auth-01	OEP-F-001	Register a new user with valid details	Portal reachable; email not already registered	1. Open /register 2. Enter name, email, password, role request 3. Submit	Riya Nair / riya@pes.edu / Pes@1234 / Student	Account created with status "pending role assignment"; confirmation shown; row present in users table	Functional	High	Aadish
TC-Auth-01a	OEP-F-001	Reject registration with a duplicate email	riya@pes.edu already registered	1. Open /register 2. Enter the same email 3. Submit	riya@pes.edu	Error "Email already registered"; no second user row created	Negative	High	Aadish
TC-Auth-01b	OEP-F-001	Reject a weak / invalid password	Portal reachable	1. Open /register 2. Enter password "123" 3. Submit	abc@pes.edu / 123	Validation error on the password field; form not submitted	Boundary	Medium	Aadish
TC-Auth-02	OEP-F-002	Login with valid credentials creates a role-based session	User exists and is approved	1. Open /login 2. Enter valid email and password 3. Submit	riya@pes.edu / Pes@1234	Redirect to the Student dashboard; JWT/session issued carrying the Student role	Functional	High	Aadish
TC-Auth-02a	OEP-F-002	Reject invalid password and invalidate session on logout	User exists	1. Login with a wrong password 2. Then login correctly, logout and replay the old token	riya@pes.edu / WrongPass1	Login fails with a generic error (no hint which field was wrong); replayed token returns 401	Negative/Security	High	Aadish
TC-Auth-03	OEP-F-003	Reset password using a valid OTP / link	Registered user; mail catcher configured	1. Click "Forgot password" 2. Enter email 3. Read OTP 4. Enter OTP and new password	riya@pes.edu, OTP from mail catcher, new password Pes@5678	Password changed; login succeeds with the new password and fails with the old one	Functional	Medium	Aadish
TC-Auth-03a	OEP-F-003	Reject an expired or incorrect OTP	OTP requested more than the validity window ago	1. Request OTP 2. Wait past expiry 3. Submit the OTP	Expired OTP 000000	Error "OTP invalid or expired"; password unchanged	Negative	Medium	Aadish
TC-Auth-04	OEP-F-004	RBAC allows each role only its permitted pages	One Student, one Instructor, one Admin exist	1. Login as each role 2. Attempt to open the dashboards of the other two roles	Student / Instructor / Admin accounts	Each role sees only its own dashboard; unauthorised navigation is blocked with an access-denied message	Functional/Security	High	Aadish
TC-Auth-04a	OEP-F-004	Direct URL/API access by an unauthorised role is blocked	Valid Student token available	1. In Postman, call an instructor-only endpoint with the Student token	GET /api/exams/manage with Student JWT	HTTP 403 Forbidden; no exam data in the response body	Negative/Security	High	Aadish

A.2 Module M2 - Exam & Question-Bank Management (Tester: Aditya Rao)

TC ID	Req ID	Test Scenario	Pre-conditions	Test Steps	Test Data	Expected Result	Type	Pri	Tester
TC-Exam-01	OEP-F-005	Create, edit and delete an exam	Logged in as Instructor	1. Open Create Exam 2. Enter title, subject, description 3. Save 4. Edit the title 5. Delete the exam	Title "DBMS Mid-Term", Subject "DBMS"	Exam is created, the edit persists after refresh, and the delete removes it from the list and the database	Functional	High	Aditya
TC-Exam-01a	OEP-F-005	Reject an exam with missing mandatory fields	Logged in as Instructor	1. Open Create Exam 2. Leave the title blank 3. Save	Empty title	Validation error; exam not created	Negative	High	Aditya
TC-Exam-02	OEP-F-006	Add MCQ, True/False and short-answer questions to the bank	Exam exists; logged in as Instructor	1. Open the question bank 2. Add one question of each type with options and the correct answer 3. Save	1 MCQ (4 options), 1 True/False, 1 short answer	All three questions are stored with the correct type, options, marks and answer key, and appear in the bank list	Functional	High	Aditya
TC-Exam-02a	OEP-F-006	Reject an MCQ saved without a correct answer	Question form open	1. Add an MCQ with options but no correct option marked 2. Save	MCQ with no key	Validation error "Select the correct option"; question not saved	Negative	High	Aditya
TC-Exam-03	OEP-F-007	Set exam schedule, duration and total marks	Exam created with questions	1. Open exam settings 2. Set start and end time, duration and total marks 3. Save	Start 10:00, End 11:00, Duration 60 min, Total 50	Schedule saved; values shown correctly on reload and used by the delivery module	Functional	High	Aditya
TC-Exam-03a	OEP-F-007	Reject an invalid schedule (end before start, or zero duration)	Exam settings open	1. Set end time earlier than start time 2. Save 3. Repeat with duration = 0	Start 11:00, End 10:00; duration 0	Validation error in both cases; invalid schedule is not persisted	Negative/Boundary	High	Aditya
TC-Exam-04	OEP-F-008	Randomise question order per student	Exam with 10+ questions; randomisation enabled; two student accounts	1. Enable shuffle 2. Login as Student A and open the exam 3. Login as Student B and open the same exam 4. Compare the order	Exam with 10 questions, two students	The two students see a different question sequence; the same set of questions and total marks	Functional	Medium	Aditya

A.3 Module M3 - Exam Delivery / Test-Taking Engine (Tester: Akshansh Agrawal)

TC ID	Req ID	Test Scenario	Pre-conditions	Test Steps	Test Data	Expected Result	Type	Pri	Tester
TC-Take-01	OEP-F-009	Student sees only eligible, in-window exams	Seeded upcoming, live and completed exams; logged in as Student	1. Open the Student dashboard 2. Review the available exams list	Live exam E1, upcoming E2, completed E3	Only E1 is startable; E2 shows as upcoming and cannot be opened; E3 shows as completed	Functional	High	Akshansh
TC-Take-01a	OEP-F-009	Exam not visible to an ineligible / unenrolled student	Student not enrolled in the exam subject	1. Login as the ineligible student 2. Check the list and try the direct exam URL	Exam E1 URL	Exam is absent from the list and the direct URL returns access denied	Negative	High	Akshansh
TC-Take-02	OEP-F-010	Countdown timer is displayed and	Live exam; Student logged in	1. Start the exam 2. Observe the timer 3.	5-minute test exam	Timer is visible and counts down; after	Functional	High	Akshansh

		enforced server-side		Refresh the page 4. Let the time run out		refresh it continues from the correct remaining time, not from the start			
TC-Take-02a	OEP-F-010	Client-side tampering with the timer does not extend the exam	Exam in progress	1. Alter the timer value in browser dev-tools 2. Attempt to submit after the real end time	Modified client timer	Server rejects the late submission or force-submits at the real end time; client value is ignored	Negative/Security	High	Akshansh
TC-Take-03	OEP-F-011	Answers auto-save periodically and survive a refresh	Exam in progress	1. Answer 3 questions 2. Wait one auto-save interval 3. Refresh the browser	3 answered MCQs	All 3 answers are restored after refresh; no data loss	Functional	High	Akshansh
TC-Take-03a	OEP-F-011	Exam auto-submits at time-up	Exam with a short duration in progress	1. Answer some questions 2. Do not submit 3. Let the timer reach zero	2-minute exam	Attempt is auto-submitted at 00:00; status becomes Submitted; answers entered so far are stored	Functional	High	Akshansh
TC-Take-04	OEP-F-012	Navigate between questions and mark for review	Exam in progress with 10 questions	1. Jump to question 7 from the navigation panel 2. Mark it for review 3. Return to question 2 4. Reopen question 7	10-question exam	Navigation works in both directions; the review flag persists and is visually distinct in the panel	Functional	Medium	Akshansh
TC-Rel-01	OEP-NF-003	99.5% availability during the simulated exam window and recovery of answers after a network disconnection	Exam in progress; test environment instrumented for uptime measurement; network throttling/disconnection available	1. Run the simulated exam window and record service uptime 2. Answer 2 questions 3. Disable the network for 30 s 4. Re-enable the network and continue 5. Submit the exam	Simulated exam window; 30 s outage; 2 answered questions	Measured availability is >= 99.5% for the simulated exam window; a reconnect indicator is shown; no answers are lost; submission completes successfully	Reliability	High	Akshansh

A.4 Module M4 - Evaluation, Results & Analytics (Tester: A S Vincent)

TC ID	Req ID	Test Scenario	Pre-conditions	Test Steps	Test Data	Expected Result	Type	Pri	Tester
TC-Eval-01	OEP-F-013	Auto-grade objective questions correctly	A submitted attempt with a known expected score	1. Submit an attempt with 7 of 10 MCQs correct 2. Open the grading result	10 MCQs x 2 marks, 7 correct	Score computed as 14/20, matching the answer key exactly	Functional	High	Vincent

TC-Eval-01a	OEP-F-013	Unanswered and partially answered questions score zero, not null	Attempt with 3 questions left blank	1. Submit the attempt 2. Inspect the per-question marks	3 blank answers	Blank answers score 0; total is arithmetically correct; no null or NaN in the scorecard	Boundary	High	Vincent
TC-Eval-02	OEP-F-014	Manual grading of subjective answers updates the total	Submitted attempt containing a short-answer question; Instructor logged in	1. Open the manual grading screen 2. Award 4 of 5 marks to the subjective answer 3. Save	Subjective answer, 4/5 marks	Marks saved; the attempt total increases by 4 and is consistent on the student scorecard	Functional	High	Vincent
TC-Eval-03	OEP-F-015	Publish results; student sees only their own scorecard	Graded attempts for two students	1. Instructor publishes the results 2. Login as Student A and view the scorecard 3. Attempt to open Student B's result URL	Student A and Student B attempts	Student A sees their own scorecard; Student B's result URL returns access denied	Functional/Security	High	Vincent
TC-Eval-03a	OEP-F-015	Unpublished results are not visible to students	Graded but unpublished exam	1. Login as Student 2. Open the results page	Unpublished exam	Message "Results not yet published"; no marks disclosed anywhere in the response	Negative	High	Vincent
TC-Eval-04	OEP-F-016	Analytics report values match the underlying data	At least 5 graded attempts exist	1. Open the analytics report 2. Compare average, pass % and item analysis against a manual calculation	5 attempts with known scores	Average, pass percentage and per-question correct % match the manual calculation exactly	Functional	Medium	Vincent

A.5 Non-Functional and Security Test Cases

TC ID	Req ID	Test Scenario	Pre-conditions	Test Steps	Test Data	Expected Result	Type	Pri	Tester
TC-Perf-01	OEP-NF-001	Exam page and navigation load within 3 s at p90	Integrated build; JMeter configured	1. Run a load profile against the exam page and question-navigation API 2. Record the latency distribution	100 virtual users, 5-minute ramp	90th-percentile response time <= 3 seconds; no errors	Performance	High	Akshansh
TC-Perf-02	OEP-NF-002	Support 500 concurrent students taking exams	Integrated build; load generator ready	1. Ramp to 500 virtual users each starting and answering an exam 2. Hold for 10 minutes 3. Check errors and data integrity	500 virtual users	No request failures, no dropped submissions; all 500 attempts recorded in the database	Scalability	High	Akshansh
TC-Sec-01	OEP-NF-004	Credentials hashed and traffic encrypted end to end	Test server with TLS enabled	1. Register a user 2. Query the users table 3. Inspect the connection in dev-tools	New user account	Stored password is a salted hash; connection uses TLS 1.2 or higher	Security	High	Vincent
TC-Sec-02	OEP-SR-001	Plain HTTP is redirected to HTTPS	TLS configured	1. Request the site over http:// 2. Observe the response	http:// URL	HTTP 301/308 redirect to https://; no application content served over plain HTTP	Security	High	Vincent
TC-Sec-03	OEP-SR-002	Passwords stored using salted	At least two users with the same	1. Create two users with identical passwords 2. Compare the stored	Two accounts, password	The two hashes differ (unique salt); neither is	Security	High	Vincent

		bcrypt/argon2	password	hashes	Pes@1234	reversible; no plaintext anywhere			
TC-Sec-04	OEP-SR-003	Server-side RBAC enforced on every protected endpoint	Valid tokens for each role	1. Call each protected endpoint with a token of the wrong role 2. Record status codes	Student, Instructor and Admin JWTs	Every mismatched call returns HTTP 403; no endpoint relies on UI hiding alone	Security	High	Vincent
TC-Sec-05	OEP-SR-004	Questions locked before start; answer key never sent to the client	Upcoming and live exams seeded	1. Request the upcoming exam before its start time 2. Start the live exam and inspect the network payload and page source	Upcoming exam E2, live exam E1	Pre-start request is refused; the live payload contains no isCorrect / correctOption field	Security	High	Vincent
TC-Sec-06	OEP-SR-005	Audit log records login, submission and grading	Audit logging enabled	1. Perform a login, an exam submission and a grading action 2. Query audit_log	One action of each type	Three records exist with actor, action, entity id and timestamp; records cannot be edited from the UI	Security	Medium	Vincent
TC-Sec-07	Cross-cutting	Input validation blocks SQL injection and XSS	Login and answer forms available	1. Submit '" OR 1=1 --' in the login field 2. Submit '<script>alert(1)</script>' as a short answer 3. Reopen the page	SQLi and XSS payload strings	Login is rejected with no SQL error exposed; the script string is escaped and displayed as text, never executed	Security	High	Vincent
TC-UX-01	OEP-NF-005	Responsive layout and WCAG 2.1 AA conformance	Build deployed; Lighthouse/axe available	1. Run an accessibility audit on the login, exam and result pages 2. Test keyboard-only navigation 3. Check 1366x768 and tablet widths	Three key pages	No critical accessibility violations; all controls reachable by keyboard; layout usable without horizontal scrolling	Usability	Medium	Akshansh

Annexure B - Defect Report Template

Every defect raised on the Team 9 Jira board must contain these fields. Tickets that omit Steps to Reproduce or Evidence are returned to the reporter (this is tracked as the defect rejection rate in Section 14.1).

Field	Description / Allowed Values	Example
Defect ID	OEP-BUG-### (auto-generated by Jira)	OEP-BUG-014
Title	One-line summary: what fails, where	Exam does not auto-submit when the timer reaches zero
Reported By / Date	Tester name and date found	Akshansh Agrawal, 26-09-2026
Module / Component	M1 / M2 / M3 / M4 / Shared	M3 - Exam Delivery
Related Requirement	Requirement ID the defect violates	OEP-F-011
Related Test Case	TC ID that detected it	TC-Take-03a
Build / Environment	Build number, branch, browser, OS	build-27, dev branch, Chrome 129, Windows 11
Severity	Critical / High / Medium / Low (see Section 14.3)	High
Priority	P1 (immediate) / P2 (this cycle) / P3 (if time permits)	P1
Pre-conditions	State required before reproducing	Student logged in; 2-minute exam in progress
Steps to Reproduce	Numbered, minimal, reproducible steps	1. Start the 2-min exam 2. Answer Q1 3. Wait for the timer to reach 00:00
Expected Result	What should happen per the acceptance criteria	Attempt is auto-submitted and marked Submitted
Actual Result	What actually happened	Timer shows 00:00 but the attempt stays In Progress and remains editable
Evidence	Screenshot, screen recording, log extract, API response	screenshot_bug014.png, server log lines 220-240
Assigned To	Developer who owns the module	Aditya Rao
Status	Open / In Progress / Fixed / Retest / Closed / Reopened / Rejected	Open
Fix Summary	Filled by the developer: root cause and change made	Timer enforcement moved from client to server-side attempt validation
Retest Result	Filled by the original reporter after the fix	Passed on build-29, 28-09-2026 - closed
Regression Impact	Test cases re-run after the fix	TC-Take-02, TC-Take-03, TC-Eval-01

B.1 Defect Lifecycle

Open -> Assigned -> In Progress -> Fixed -> Retest (by the original reporter) -> Closed. If the retest fails, the defect moves to Reopened and returns to the developer. A defect that is not a genuine fault is moved to Rejected with a written justification and is reviewed at the next triage meeting.

Annexure C - Requirements Traceability Matrix (RTM)

All 26 requirements from the SRS traced to design specification, owning module, test case(s), responsible tester and status. Status codes: N = Not started, P = Partial, A = Accepted. This matrix is updated after every test cycle and must show all rows as A before sign-off.

Req ID	Requirement (short)	Section / Design Spec	Module	Test Case(s)	Tester	Status
OEP-F-001	Register user	4.1 / DS-Auth-01	M1 AuthModule	TC-Auth-01, 01a, 01b	Aadish	N
OEP-F-002	Login and role-based session	4.1 / DS-Auth-02	M1 AuthModule	TC-Auth-02, 02a	Aadish	N
OEP-F-003	Password reset via OTP	4.1 / DS-Auth-03	M1 AuthModule	TC-Auth-03, 03a	Aadish	N
OEP-F-004	Role-based access control	4.1 / DS-Auth-04	M1 AuthModule	TC-Auth-04, 04a, TC-Sec-04	Aadish	N
OEP-F-005	Create / edit / delete exam	4.2 / DS-Exam-01	M2 ExamModule	TC-Exam-01, 01a	Aditya	N
OEP-F-006	Question bank (MCQ/TF/short)	4.2 / DS-Exam-02	M2 ExamModule	TC-Exam-02, 02a	Aditya	N
OEP-F-007	Schedule, duration, total marks	4.2 / DS-Exam-03	M2 ExamModule	TC-Exam-03, 03a	Aditya	N
OEP-F-008	Randomise question order	4.2 / DS-Exam-04	M2 ExamModule	TC-Exam-04	Aditya	N
OEP-F-009	List available exams	4.3 / DS-Take-01	M3 DeliveryModule	TC-Take-01, 01a	Akshansh	N
OEP-F-010	Timed delivery with countdown	4.3 / DS-Take-02	M3 DeliveryModule	TC-Take-02, 02a	Akshansh	N
OEP-F-011	Auto-save and auto-submit	4.3 / DS-Take-03	M3 DeliveryModule	TC-Take-03, 03a, TC-Rel-01	Akshansh	N
OEP-F-012	Navigation and mark-for-review	4.3 / DS-Take-04	M3 DeliveryModule	TC-Take-04	Akshansh	N
OEP-F-013	Auto-grade objective questions	4.4 / DS-Eval-01	M4 EvalModule	TC-Eval-01, 01a	Vincent	N
OEP-F-014	Manual grading of subjective	4.4 / DS-Eval-02	M4 EvalModule	TC-Eval-02	Vincent	N
OEP-F-015	Publish and view results	4.4 / DS-Eval-03	M4 EvalModule	TC-Eval-03, 03a	Vincent	N
OEP-F-016	Analytics and reports	4.4 / DS-Eval-04	M4 EvalModule	TC-Eval-04	Vincent	N
OEP-NF-001	Response time <= 3 s (p90)	5 / DS-Perf-01	All modules	TC-Perf-01	Akshansh	N
OEP-NF-002	500 concurrent students	5 / DS-Perf-02	All modules	TC-Perf-02	Akshansh	N
OEP-NF-003	99.5% availability / recovery	5 / DS-Rel-01	M3 DeliveryModule	TC-Rel-01	Akshansh	N
OEP-NF-004	Hashing and TLS 1.2+	5 / DS-Sec-01	Shared / M1	TC-Sec-01, TC-Sec-03	Vincent	N
OEP-NF-005	Responsive UI, WCAG 2.1 AA	5 / DS-UX-01	All UI	TC-UX-01	Akshansh	N
OEP-SR-001	TLS mandatory, HTTP redirected	5.1.2 / DS-Sec-02	Shared	TC-Sec-02	Vincent	N
OEP-SR-002	Salted bcrypt/argon2 hashing	5.1.2 / DS-Sec-01	M1 AuthModule	TC-Sec-03	Vincent	N
OEP-SR-003	Server-side RBAC on every API	5.1.2 / DS-Sec-03	M1 + all APIs	TC-Sec-04, TC-Auth-04a	Vincent	N
OEP-SR-004	No pre-start exposure / no answer key in payload	5.1.2 / DS-Sec-04	M2 + M3	TC-Sec-05	Vincent	N
OEP-SR-005	Audit log of login, submit, grade	5.1.2 / DS-Sec-05	Shared	TC-Sec-06	Vincent	N
Cross-cutting test item (not an SRS requirement)	Input validation (SQLi / XSS)	5.1 / DS-Sec-06	All input paths	TC-Sec-07	Vincent	N

Coverage summary: 26 requirements, 100% traced to at least one test case; 40 test cases total (24 functional positive, 11 negative/boundary, 10 security, 2 performance, 1 reliability, 1 usability - some cases cover more than one category).